

Student t distribution

In z stats when we perform any analysis using z-score
we require σ (population standard deviation) $\rightarrow$ is already known

Also every time having population S.D. is very difficult bcs we don't have every time entire population data

How do we perform any analysis when we don't know
the population standard deviation?



Student's t distribution

t stats

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Z table

converted to...

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

S = Sample Standard deviation

t table $\Rightarrow$ t test

Before we are looking at z-table now we are looking at the t-table

In the t-table first have to look for the tails option like one tail or two tail
then
see for df = degree of freedom

Degree of freedom

$$\text{dof} = n - 1 = 3 - 1 = \underline{\underline{2}}$$

3 people



chairs

When first person come to the room he has three options
when 2nd person come he has now only 2 options and
when 3rd person comes he not have any options